Shrinkage Estimator Based Regularization for EEG Motor Imagery Classification

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***Abstract*—Electroencephalography (EEG) based motor imagery Brain-Computer Interface (MI-BCI) paradigm is used to communicate with external device by people who lost peripheral nerve control, or perform neuro-rehabilitation for stroke patients. BCI systems based on motor imagery often employ feature extraction algorithms based on Common Spatial Patterns (CSP). CSP is capable of discriminating two classes, but it is sensitive to outliers and noisy trials. Therefore, regularisation is often deployed to improve the robustness and accuracy of CSP estimation. In this paper, a novel regularisation approach based on shrinkage estimation is presented in order to handle small sample problem and retain subject-specific discriminative features. In this method, an analytical solution for shrinkage estimation is provided, which not only is computationally tractable, but also overcomes the heuristic approach of traditional cross validation based parameter tuning. We applied the proposed regularisation to Filter-Bank Common Spatial Pattern (FBCSP). The proposed method is evaluated on two publicly available datasets, namely Wadsworth Physiobank Dataset and BCI Competition IV Dataset 2a. The results show that Shrinkage Regularized Filter Bank CSP (SR- FBCSP) outperforms FBCSP in classifying left vs right hand motor imagery.**

***Index Terms*—Brain Computer Interface, EEG, regularization, shrinkage estimation, motor imagery, Common Spatial Pattern**

I. Introduction

Brain Computer Interface (BCI) is garnering widespread attention among researchers from a multitude of disciplines like signal processing, machine learning and neuroscience [1]. BCI is a technology which records the neural information associated with mental tasks and bypasses the conventional neuromuscular pathway to communicate with the external world. As a non-invasive platform to capture brain activities, Electroencephalography (EEG) is one of the most widely used BCI modalities because of its high temporal resolution, portability and cost-effectiveness. In particular, EEG based Motor imagery (MI) BCI has become very popular in the area of Neurorehabilitation and neuroprosthesis as it involves decoding the movement intentions to help stroke patients and amputees [2], [3].

Since the practical settings of Neurorehabilitation require real-time MI decoding, it is very important to achieve high efficacy with lesser computational expenses. Especially in an online scenario which involves single trial EEG analysis, it is equally important to address the inter-trial variability and low signal-to-noise ratio (SNR) [4]. In addition to that, it is essential to characterize the spatiotemporal activity as precise as possible. In this regard, EEG inverse modeling assists us to locate the cortical sources responsible for a given mental task [5]. However, EEG source localization is an ill-posed problem due to the cortical sources vastly outnumbering the scalp electrodes (channels) used to record the neural activity. Especially the background neural activity contributes to the spatiotemporal noise which affects the signal covariance matrix which is one of the most important a priori information required for EEG source localization. Also, it is to be noted that the covariance matrix estimation plays a pivotal role in Common spatial patterns (CSP) - an algorithm which is very popular in EEG based motor imagery/execution classification. On the downside, CSP is not very robust when the training sample size is not big enough. Therefore, some regularization approach is required which is robust to small sample settings as well as outliers.

Although there have been some studies which regularize the covariance matrix based on weighted approach, most of them depend on either hand-picked constant or cross-validation [6]. However, for online settings it is important to find an analytical solution for regularization parameter which is not very computationally intensive. The objective of this regularization approach is to be able to handle trial-to-trial variability, which is inherent in most of the EEG based BCI experiments. Therefore, shrinkage estimator based regularized approach for filter bank common spatial pattern is presented in this study. In the context of EEG covariance matrix estimation, shrinkage parameter has been used to analyse normalized distance to true covariance matrix [7]. The novelty of our paper is, to use this estimator as a regularization technique for CSP to classify motor imagery EEG. It also has the advantage of an analytical solution for the regularization parameter and does not require heuristic evaluation, thus very useful in an online BCI. This estimator regularizes the sample-to-sample variables in the empirical covariance matrix, thus resulting in more robust estimation. Two datasets of considerably different number of subjects, channels and trials are evaluated in this study. The filter bank approach is preferred due to the discriminative subject-specific frequency range, which helps in improving the classification accuracy [8], [9]. Also, this paper investigates the performance of regularized approach using mutual information based feature selection technique known as "minimum redundancy maximum relevancy" (mRMR).

This paper is organized as follows: Section II describes the methodology in which theoretical framework for shrinkage regularized Common Spatial Pattern (SR-CSP) and Shrinkage regularized Filter Bank Common Spatial Pattern (SR-FBCSP) is presented. Furthermore, Section III describes the dataset used in this study. Section IV presents the evaluation results and section V concludes the paper.

II. Methodology

The proposed methodology involves zero-phase shift bandpass filtering of raw EEG signal between 4-40Hz using 5th order Butterworth filter. Since the proposed method focuses on filter bank Common Spatial Pattern, 9 different filter banks of 4Hz bandwidth each i.e. 4-8Hz, 8-12Hz, 12-16Hz,..., 36-40Hz is used in this study. Filtered signal is then segmented into epochs based on the trial label and duration. Subsequently, these filtered signals are projected into a different subspace using shrinkage regularized Common Spatial pattern. New feature space is minimized using a feature selection algorithm called minimum redundancy maximum relevancy (mRMR) [10]. For further feature classification, Support Vector Machine Classifier based on LIBSVM package [11] is used. 10-fold cross validation is performed to ensure robustness.

A. Shrinkage estimator based Regularized Common Spatial Pattern (SR-CSP):

Common Spatial Pattern (CSP) is one of the most widely used algorithms in BCI experiments because of its capability in discriminating two classes very effectively [12]. However, the CSP has its own limitations of being highly sensitive to spatiotemporal noise and severely overfit in the cases where the training set is small [13]. CSP does a linear transformation of EEG signal (*N* channels $\times$ *T* samples per trial) into a low-dimensional subspace. The rows of resultant projection matrix 'W' consist of weights for each channel. This transformation basically maximizes the inter-class variance and minimizes the intra-class variance of 2 classes and by simultaneously diagonalizing the covariance matrices and which are expressed as shown in (1) [12],

$$C_{\mathrm{E}_i} = \frac{E_i E_i^T}{trace(E_i E_i^T)} \quad \forall i = \{1, 2\} \tag{1}$$

where, E_i denotes $E \in \mathbb{R}^{N \times T}$ for class '*i*' . *Trace*(*E*) gives the sum of all the diagonal elements in the matrix '*E*'. In conventional methods, these spatial covariance matrices C_{E_1} and C_{E_2} are averaged over the trials to give sample covariance matrices S_1 and S_2 which are denoted as (2),

$$S_1 = \frac{1}{N_1}\sum_{j=1}^{j=N_1} C_{\mathrm{E}_1}(j) \quad S_2 = \frac{1}{N_2}\sum_{j=1}^{j=N_2} C_{\mathrm{E}_2}(j) \tag{2}$$

where, N_1 and N_2 represents the number of class 1 and class 2 trials respectively.

Most of the conventional techniques for covariance estimation employ sample covariance estimator. However, it is to be noted that an estimator which is both well-conditioned and more accurate than sample covariance matrix is needed to achieve robust performance in small-sample settings [14], [15]. This estimator is the weighted average of sample covariance matrix and the structured estimator. Let us say, Ω is the unknown true covariance matrix and $\hat{\Omega}_S$ is a structured estimator which minimizes the least square error i.e., LSE ($\hat{\Omega}_S$) $= \min(E\{\left\|\hat{\Omega}_S - \Omega\right\|_F^2\})$, (where $\|.\|_F$ is the Frobenius norm), then the shrinkage estimator allows us to find the optimal shrinkage parameter (α_Ω). The idea is to use shrinkage estimator instead of the sample covariance matrix (S_Ω) for eigenvalue decomposition in Common spatial pattern algorithm.

$$\hat{\Omega}_S = \alpha_\Omega T_\Omega + (1 - \alpha_\Omega) S_\Omega \tag{3}$$

Although there are studies in which structured estimator is used to regularize CSP [16], [17], the regularization parameter is chosen by performing cross-validation. However for online settings, an analytical solution is preferred for regularization. To find an analytical solution for the shrinkage parameter, minimization of least square error is performed as mentioned earlier.

$$\begin{aligned} LSE(\hat{\Omega}_S) &= \mathbb{E}\{\left\|\hat{\Omega}_S - \Omega\right\|_F^2\} \\ &= \mathbb{E}\{\|\alpha_\Omega(T_\Omega - S_\Omega) + (S_\Omega - \Omega)\|_F^2\} \\ &= \alpha_\Omega^2 \mathbb{E}\{\|T_\Omega - S_\Omega\|_F^2\} + \mathbb{E}\{\|S_\Omega - \Omega\|_F^2\} \\ &\quad + 2\alpha_\Omega \mathbb{E}\{tr\{(T_\Omega - S_\Omega)(S_\Omega - \Omega)^T\}\} \end{aligned} \tag{4}$$

Here, the relation $\|A + B\|_F^2 = \|A\|_F^2 + \|B\|_F^2 + 2\mathrm{tr}\{AB^T\}$ for square matrices is used. Shrinkage parameter α_Ω is derived by equating the first order differential of LSE($\hat{\Omega}_S$) to zero.

$$\hat{\alpha}_\Omega = \frac{\mathbb{E}\{tr\{(T_\Omega - S_\Omega)(S_\Omega - \Omega)^T\}\}}{\mathbb{E}\{\|(T_\Omega - S_\Omega)\|_F^2\}} \tag{5}$$

The numerator can be decomposed into ,

$$\begin{aligned}
&\mathbb{E}\{tr\{(S_\Omega - T_\Omega)(S_\Omega - \Omega)^T\}\} \\
&\quad = \mathbb{E}\{\|S_\Omega\|_F^2\} - tr\{\mathbb{E}\{T_\Omega S_\Omega^T\}\} \\
&\qquad - tr\{\mathbb{E}\{S_\Omega - T_\Omega\}\Omega^T\} \\
&\quad = \mathbb{E}\{\|S_\Omega\|_F^2\} - tr\{\mathbb{E}\{T_\Omega S_\Omega^T\}\} \\
&\qquad - tr\{\mathbb{E}\{S_\Omega - T_\Omega\}E\{\Omega^T\}\} \\
&\qquad + tr\{\mathbb{B}\{S_\Omega - T_\Omega\}\{\Omega^T\}\} \\
&\quad = \underbrace{\mathbb{E}\{\|S_\Omega\|_F^2\} - \|\mathbb{E}\{S_\Omega\}\|_F^2}_{\mathbb{V}\{S_\Omega\}} \\
&\qquad - \underbrace{tr\{\mathbb{E}\{T_\Omega S_\Omega^T\}\} - \mathbb{E}\{T_\Omega\}\mathbb{E}\{S_\Omega^T\}\}}_{\mathbb{C}\{S_\Omega, T_\Omega\}} \\
&\qquad + tr\{\mathbb{E}\{S_\Omega - T_\Omega\}\{\Omega^T\}\}
\end{aligned} \tag{6}$$

If we consider S_Ω as an unbiased estimator of Ω, then $\mathbb{B}\{\Omega^T\}$ in the above expression becomes zero. Therefore, the analytical expression for α_Ω is given as,

$$\hat{\alpha}_\Omega = \frac{\mathbb{V}\{S_\Omega\} - \mathbb{C}\{S_\Omega, T_\Omega\}}{\mathbb{E}\{\|(T_\Omega - S_\Omega)\|_F^2\}} \tag{7}$$

The composite spatial covariance matrix C_c expressed as,

$$C_C = \bar{\Omega}_{S_1} + \bar{\Omega}_{S_2} \tag{8}$$

In the above equation (8), $\bar{\Omega}_{S_i}$ (where, $i = \{1, 2\}$) denotes the shrinkage estimator based covariance matrix (3) averaged over all the trials corresponding to class 'i'.

The resultant spatial covariance matrix C_c can be modelled as a generalized eigenvalue problem $C_c = U_c \lambda_{\mathbf{c}} U_c^T$, where U_c is the matrix of eigenvectors and λ_c is the diagonal matrix of eigenvalues. This singular value decomposition problem can be solved using a whitening transformation $P = \sqrt{\lambda_{\mathbf{c}}^{\mathbf{-1}}} U_c^T$. This results in new matrices Σ_1 and Σ_2 which share common eigenvectors *P* represented as shown in (9),

$$\Sigma_1 = P\bar{\Omega}_{S_1}P^T \qquad \Sigma_2 = P\bar{\Omega}_{S_2}P^T \tag{9}$$

This transformation can be alternatively represented as $\Sigma_1 = \Gamma\mathbf{\Lambda}_1\Gamma^T, \Sigma_2 = \Gamma\mathbf{\Lambda}_2\Gamma^T$. The sum of corresponding eigenvalues is equated to an identity matrix (10).

$$\mathbf{\Lambda}_1 + \mathbf{\Lambda}_2 = \mathrm{I} \tag{10}$$

Since the above property always holds true, the maximum eigenvalue of Σ_1 corresponds to the minimum eigenvalue of Σ_2. This constrained property makes the eigenvectors to discriminate two classes more effectively. Now the EEG data *E* is projected into a new spatial subspace **Z** with *W* being the projection matrix, which is written as (12),

$$W = ({U_c}^T P)^T \tag{11}$$

$$\mathbf{Z} = W \times E \tag{12}$$

Features are log transformed normalized variance of Z_p as shown in (13),

$$F_p = log\left[var(\mathbf{Z}_p) / \sum_{i=1}^{i=2m} var(\mathbf{Z}_p)\right] \tag{13}$$

B. Shrinkage Estimator based Regularized Filter Bank Common Spatial Pattern (SR-FBCSP):

The mathematical framework for Shrinkage estimator based Regularized Filter bank Common Spatial Pattern (SR-FBCSP) is similar to that of SR-CSP, but differs in terms of number of output features. FBCSP involves decomposition of EEG signal $E \in \mathbb{R}^{N \times T}$ into multiple filter banks ranging from theta to gamma frequency bands i.e., $E = \{E_{4-8Hz}, E_{8-12Hz}, ..., E_{36-40Hz}\}$. Each filter bank results in corresponding projection matrices $W = \{W_1, W_2, ..., W_9\}$. Resulting feature space is reduced by applying feature selection method called "minimum redundancy maximum relevancy" (mRMR). It takes into account the mutual information between the class labels c_i and the individual features x_i and are ranked accordingly. In a given feature space, it is possible that some of the features have dependance on each other and their combination would be redundant as only one of the feature would alone be sufficient to contribute to the accuracy. In this regard, it is necessary to have only those features which are actually relevant. Mathematically, the maximization of distance between relevancy and redundancy of features is expressed as in (14) [10],

$$\max \xi(D, R), \qquad \xi = D - R \tag{14}$$

In (14), D and R denotes the relevancy and redundancy respectively, which are the functions of mutual information between a feature and a class label [10].

III. Evaluation data

To evaluate the performance of the proposed variants of CSP, EEG data from 2 datasets namely Wadsworth physiobank data and BCI Competition IV Dataset 2a are used in this study. In the following sections, these data will be referred to as *Dataset-1* and *Dataset-2* respectively.

A. Dataset-1

Wadsworth physiobank dataset [18], [19] has EEG data from 109 subjects performing tasks like right fist, left fist motor imagery and motor execution, both fists and feet motor imagery and motor execution. However, only the EEG trials corresponding to right fist and left fist motor imagery are used in this study. EEGs are recorded from 64 channels which follow the International 10-10 system. Out of the 14 experimental runs, only 3 runs numbering 4, 8 and 12 corresponding to right fist and left fist MI are evaluated. Each of these 3 MI runs have 15 trials each, thus collectively resulting in 45 trials per subject. Each trial (either right fist MI or left fist MI) would last for 4.1s followed by a rest period of 4s and the sampling rate is 160 Hz. Some of the subjects had bad annotation files due to which, 85 subjects are evaluated in this study.

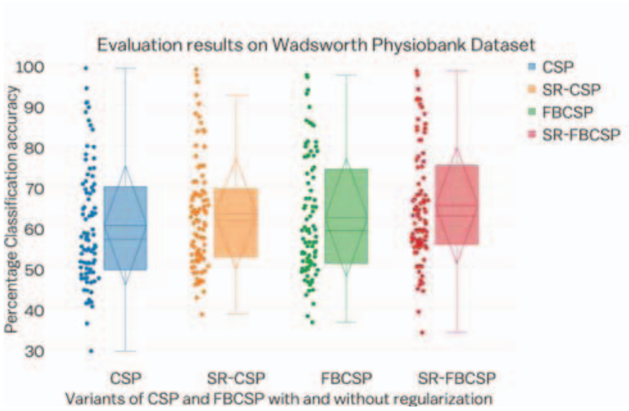

Fig. 1: Boxplot of 10 $\times$ 10 cross validation accuracies obtained on Dataset-1 using variants of CSP and FBCSP with and without regularization

B. Dataset-2

BCI Competition IV Dataset 2a, provided by the Graz group has 4-class EEG data recorded from 9 subjects [20]. 4-class data corresponds to left hand, right hand, foot and tongue motor imagery. 22 Ag/AgCl electrodes with inter-electrode distance of 3.5 cm are used to record the EEG signal. Out of 4-class EEG data, only the trials corresponding to left hand and right hand motor imagery are considered in this study. A total of 144 trials corresponding to both the classes is evaluated.

IV. Results and Discussion

The performance of the proposed method is evaluated based on the classification accuracy as resulted by 10 fold cross validation using the SVM classifier. Since Dataset-1 used in this study has 85 subjects, subject-wise accuracy is difficult to interpret. Therefore, a jittered boxplot is shown in Fig. 1 in which the jittered data points show the distribution of classification accuracies of various subjects and the boxplot itself highlights information regarding the mean and median classification accuracies. From Fig. 1, it can be observed that SR-FBCSP and SR-CSP outperforms FBCSP and CSP by atleast 3%. Also, SR-FBCSP performs better compared to SR-CSP due to discriminative information from multiple frequency bands. These results are in agreement with [21], which presents superiority of FBCSP compared to CSP. In [21], FBCSP is integrated with Mutual Information based Best Individual feature (MIBIF) as its feature selection algorithm and Naive Bayesian Parsen Window (NBPW) as its classification technique. In the current work, FBCSP is integrated with mRMR and SVM for feature selection and classification. It can be observed that the results for Dataset-2 shows slightly improved accuracy compared to [21].

It is to be noted that the number of training trials is very less in Dataset-1 compared to that of Dataset-2. Therefore, 10 $\times$ 10 cross validation is performed to ensure the same training to testing ratio. From the results shown in Fig 1 and Fig 2, it can be observed that regularized approach (SR-CSP and SR-FBCSP) performs much better than CSP and FBCSP in Dataset-1, however it is not so significant in Dataset-2. Reason for this can be attributed to the fact that shrinkage regularized approach makes a difference when the training set is small. Otherwise the shrinkage parameter (α_Ω) in (3) approaches zero, thus reducing the whole expression Ω_S into S_Ω which is essentially the sample covariance matrix.

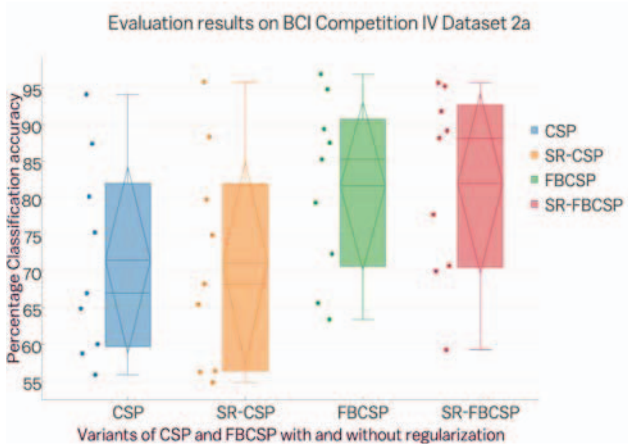

Fig. 2: Boxplot of 10 $\times$ 10 cross validation accuracies obtained on Dataset-2 using variants of CSP and FBCSP with and without regularization

Apart from the classification performance distribution shown in Fig 2, subject-specific classification accuracies for Dataset-2 is shown in Table I. From this table, an improvement of slightly more than 10% can be observed when the filter bank approach is used. Also, our proposed SR-FBCSP method results in better accuracy, if not statistically significant compared to the regularized approaches presented in [22]. In addition to this, it is interesting to observe the spatial patterns confirming the neurophysiological correlation of left and right hand motor imagery in Fig 3. In this figure, the topographical plot of SR-FBCSP is shown for few representative subjects from Dataset-1. It can be observed that there is a clear discrimination between left and right hand motor imagery in its contralateral brain region, thus validating the neurophysiological plausibility of the proposed method.

TABLE I: Percentage Classification accuracies obtained for each subject of Dataset-2 for the standard CSP and its regularized variants

Subjects	CSP(%)	SR-CSP (%)	FBCSP (%)	SR-FBCSP (%)
A01	56.38	58.75	87.51	89.24
A02	68.21	66.99	63.41	59.36
A03	79.75	80.17	96.87	95.81
A04	54.79	55.84	72.34	70.73
A05	65.42	64.87	79.34	78.1
A06	56.25	60.02	65.64	69.98
A07	74.88	75.27	85.26	88.24
A08	95.77	94.11	94.8	95.32
A09	88.33	87.36	89.39	91.92
Mean	71.0867	71.4867	81.6178	82.0656

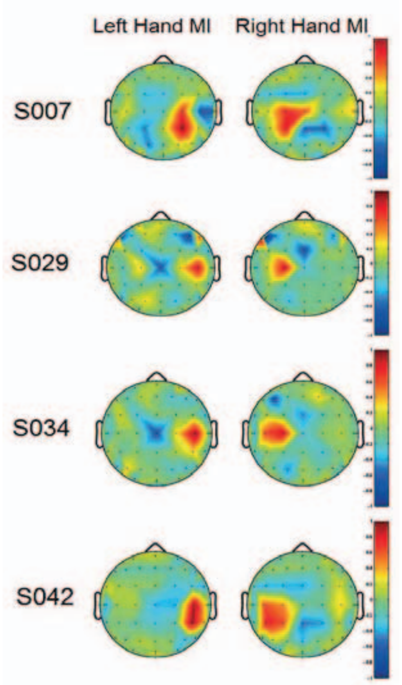

Fig. 3: Visualization of regularized filter bank Common Spatial patterns in some representative subjects of Dataset-1

CONCLUSION

In this paper, a shrinkage parameter based regularized approach for Filter bank Common Spatial Pattern is presented. The proposed method is evaluated on EEG data from 2 publicly available datasets. Overall, Shrinkage Regularized Filter Bank Common Spatial Pattern (SR-FBCSP) gave better results compared to other variants of CSP. For a bigger dataset, shrinkage parameter approaches zero, thus resulting in a similar performance as that of a non-regularized approach. On the contrary, SR-FBCSP gives better results when the number of training trials is very less. This is validated from the results obtained on Dataset-1 with less than 25 trials per class for a given subject. The results are motivating enough to employ the proposed method in online BCI with very few training samples to reduce the calibration time. Furthermore, this shrinkage approach for covariance estimation plays an important role in the EEG source localization, which will be investigated in the future work. It would be interesting to observe the effect of regularized spatial covariance matrix on inverse modeling for EEG source localization to analyse single-trial EEG.

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